# Le Nguyen Gia Hung

AI undergraduate | Machine learning, computer vision & speech

First & corresponding author of a published AJCAI 2025 paper.

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## Education

**FPT University** — B.Eng. in Artificial Intelligence Expected Dec 2027
GPA: **8.62/10.0**. Certificate of Merit: semesters 3, 4, 5 and 6.

## Technical Skills

**Languages & data:** Python, SQL/PostgreSQL, Pandas, NumPy.
**Machine learning:** PyTorch, Transformers, TensorFlow/Keras, Scikit-learn, OpenCV.
**Evaluation & tools:** SHAP, Integrated Gradients, DiCE, chronological cross-validation; Git, Docker, Linux, Weights & Biases, Streamlit, FastAPI.

## Selected Research & Projects

**SmokeNet: Air Quality Forecasting** Jun – Nov 2025
*First & Corresponding Author; published at AJCAI 2025*

- Designed a Transformer to forecast $PM_{2.5}$; used seasonal embeddings, reconstruction loss and chronological evaluation on **4+ years** of OpenWeather data.
- Reported **MAE 0.7470**, $R^2$ **0.9545** and **35.9% lower MAE than XGBoost** on the 2024 held-out evaluation. Inspected forecasts with SHAP, Integrated Gradients and DiCE. Case study.

**MERR-GAT: Multimodal Emotion Recognition** Oct 2025 – Present
*Second & Corresponding Author; research in progress*

- Develop emotion recognition from speech and text with **WavLM + Compact Conformer**, BERT/RoBERTa and dual-channel graph-attention fusion.
- Evaluate on IEMOCAP, RAVDESS and MELD using WA, UA and Macro-F1; experiments are ongoing. Methods and code links.

**Vietnamese Speech Recognition** Sep 2025 – Present
*Co-Researcher & Co-Author; research in progress*

- Develop a **5.4M-parameter** CNN–residual BiLSTM with CTC on **745 hours** of Vietnamese speech; reported **WER 33.09%**, **CER 15.28%**. Compare with PhoWhisper. Evaluation notes.

**PPE Detection** — YOLOv8, OpenCV, Streamlit Course project

- Built **9-class** detection, safe/unsafe compliance logic and a webcam interface with violation reporting. Project details.

## Leadership & Cloud Learning

**SpeedyLabX, FPT University** Jan 2025 – Present
*Founder & Student Lead*

- Lead a **10-member** undergraduate group with weekly technical workshops, paper reading and shared conference-submission workflows.

**First Cloud AI Journey** — FCAJ Cloud Intern, Team NeuraX 2026
AWS Vietnam program; cloud learning and project work documented in the program journal.

## Publication

**Gia-Hung Nguyen Le** et al. "Proactive Air Quality Forecasting and Health Alert System for Melbourne." *AJCAI 2025, Springer LNAI*. DOI: 10.1007/978-981-95-4969-6_34.
Three additional manuscripts under review; full titles and contributions in the research CV.